ECE 501: Digital Image Processing

Course Project – Week 7 Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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Project 17, Group 7

Project Title

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Team Details

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1 Work Done This Week

1.1 Starting Point and Goals

This report covers only work done after the Week 6 report and does not repeat its results or definitions. Week 6 left three problems open: false alarms from white margins and fabric edges, tiny defects that patch statistics barely show, and results spread over several defect-free samples. This week we (i) removed margin false alarms, (ii) added features aimed at thin defects, (iii) tested smaller patches, and (iv) compared all variants on the frozen sample with one set of measures and confidence intervals, so that the final configuration could be chosen by a rule fixed in advance.

1.2 One Frozen Data Sample

All results use Sample C of the Week 6 report (the 103 mask-matched defective and all 141 defect-free images, 20 or 21 per fabric) and the same setup (per-fabric models and thresholds, out-of-fold scores from image-level cross-validation). Baseline A is the best Week 6 configuration (Gabor + FFT, CLAHE + blur, per-fabric model) recomputed on this sample, so its values differ from the Sample A values of Week 6 (defect recall 0.196 and 0.431 against 0.175 and 0.379 at 1% and 5% FPR), and its image-level AUC (0.766) is the Sample C value already reported there. The features recomputed by this week's code reproduce the cached ones exactly.

1.3 Margin Mask: A Mistake and Its Correction

White margins and the fabric edge next to them produce false alarms, so we excluded them from training, thresholds, scoring and evaluation ("margin mask"). Our first rule masked every patch with at least 2% pure-white pixels. The example figures showed that this rule also masked *bright (white) defects*: in one missed example the mask covered the white defect itself, and the detection rate of defect type 006 (white defects) fell to 12.5% at 1% and 5% false-positive rate (FPR), against 100% for the unmasked baseline in the Week 6 report. The masking statistics had been a warning too: 236 of 244 images had a masked zone and 15.3% of the defect-overlapping patches (320 of 2,087) were lost, although only some images have real white margins. We corrected the rule: a margin is a run of at least 48 consecutive image columns in which at least 80% of the pixels are white (a wide, full-height blank band), widened by 48 px on each side to cover the fabric edge. Thin bright defects, which are only a few pixels wide, are never masked. With the corrected rule 11.5% of all 32×32 patches are masked and 0.8% of the defect-overlapping patches (41 of 5,080) are lost. All masked results below use the corrected rule.

1.4 Thin-Defect Features

Most defects are only a few pixels wide, so patch means and variances of the Gabor responses dilute them. For 64×64 patches we added ten features to the existing 24 Gabor and FFT features: the 95th percentile of the absolute Gabor response in the patch for each of the eight Gabor filters, and the spread (standard deviation) of the patch's column means and of its row means, which react to line-shaped structures. This gives 34 features per patch.

1.5 Smaller Patches

We also tested 32×32 patches with stride 16 (933,015 patches, of which 5,080 overlap a defect, against 216,846 patches at 64 px). Each patch has 34 features: for each Gabor map the mean, the variance and the maximum absolute response (24), four radial FFT bands with mean and standard deviation (8), and the column and row profile spreads (2). The window statistics are computed from the filtered whole image, which is fast; we verified that they equal the statistics computed directly on the patch windows.

1.6 Six Configurations and One Evaluation Protocol

We compared six configurations: **A** the baseline (64 px, 24 features, no mask); **B** A with the margin mask; **C** 64 px with the thin-defect features; **D** C with the mask; **E** 32 px patches; **F** E with the mask. The measures are those defined in Week 6 (defect recall and false alarms, Section 1.4; the top-k localization test, Section 1.8; image-level AUC and recall at fixed FPR, Section 1.9). What is new this week:

- For 32 px patches the localization test uses the top-40 patches, which cover the same area as the top-10 patches at 64 px.
- False-alarm *counts* are not comparable between patch sizes: a 32 px patch has a quarter of the area and there are about 4.3 times as many patches, while the false-alarm *area* is similar (33.6 patches of 1,024 px against 7.8 patches of 4,096 px per image at 1% FPR). We therefore do not rank on this count.
- Confidence intervals come from a paired image bootstrap (500 resamples, the same resamples for every configuration). The decision rule was fixed before looking at the results: adopt a change only if at least two measures improve with an interval that excludes zero and none gets reliably worse.
- The 32 px configurations were run with one cross-validation seed.

2 Results

2.1 Comparison of the Six Configurations

Table 1: Six configurations on all 103 defective and 141 defect-free images; brackets: 95% CI from resampling images. Top-10 hit rate: top-10 patches for 64 px, top-40 for 32 px (equal area). False-alarm counts are not comparable between patch sizes.

Configuration	Recall 1% FPR	Recall 5% FPR	False alarms/img (1%)	Top-1 hit	Top-10 hit	Image AUC	Image recall 10% FPR
A baseline (64 px)	0.196 [0.119, 0.276]	0.431 [0.333, 0.527]	8.8	0.118 [0.059, 0.184]	0.225 [0.148, 0.311]	0.766 [0.693, 0.823]	0.515
B A + mask	0.265 [0.179, 0.350]	0.451 [0.353, 0.539]	7.8	0.196 [0.119, 0.275]	0.353 [0.257, 0.451]	0.767 [0.693, 0.822]	0.505
C thin features	0.196 [0.122, 0.275]	0.431 [0.338, 0.525]	8.8	0.137 [0.078, 0.205]	0.216 [0.136, 0.294]	**0.802** [0.735, 0.852]	0.515
D C + mask	0.255 [0.169, 0.330]	0.451 [0.350, 0.539]	7.8	0.245 [0.162, 0.332]	0.392 [0.294, 0.485]	0.782 [0.716, 0.838]	0.534
E 32 px	0.314 [0.223, 0.398]	0.735 [0.641, 0.816]	38.0	0.167 [0.098, 0.245]	0.441 [0.343, 0.544]	0.772 [0.707, 0.827]	0.553
F 32 px + mask	**0.451** [0.353, 0.542]	**0.745** [0.650, 0.823]	33.6	**0.304** [0.214, 0.402]	**0.510** [0.408, 0.608]	0.772 [0.703, 0.828]	0.573

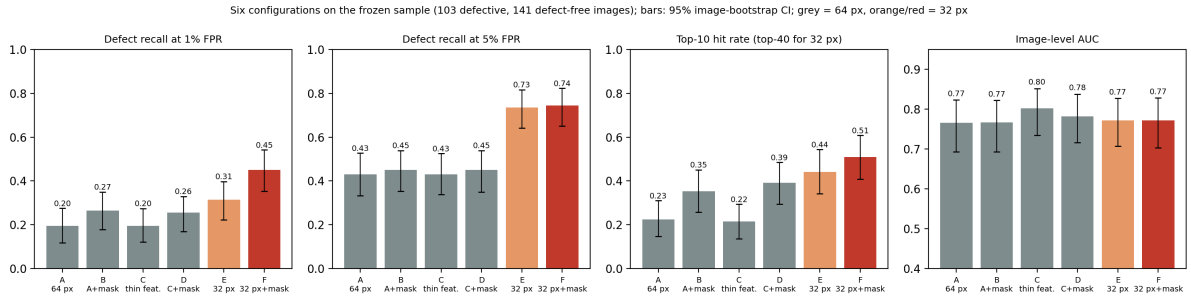


Figure 1: Four measures for the six configurations (95% image-bootstrap intervals). Smaller patches raise recall at 5% FPR from about 0.43 to 0.74; the margin mask helps localization and recall at 1% FPR; image-level AUC is best with the 64 px thin-defect features.

2.2 Paired Differences

Table 2: Differences between configurations (median and 95% CI over paired image-bootstrap resamples; $*$ = interval excludes 0). Differences of B and D to A were not tested separately after the mask correction.

Configuration	vs	Recall 1% FPR	Recall 5% FPR	Top-1 hit	Top-10 hit	Image AUC
C thin features	A	0.000 [−0.039, +0.030]	0.000 [−0.079, +0.068]	+0.019 [−0.020, +0.059]	−0.010 [−0.058, +0.029]	+0.036 [+0.010, +0.064] $*$
E 32 px	A	+0.117 [+0.058, +0.188] $*$	+0.303 [+0.214, +0.404] $*$	+0.049 [+0.010, +0.094] $*$	+0.214 [+0.138, +0.294] $*$	+0.007 [−0.031, +0.041]
E 32 px	D	+0.059 [0.000, +0.127]	+0.282 [+0.197, +0.376] $*$	−0.078 [−0.139, −0.020] $*$	+0.040 [−0.020, +0.109]	−0.011 [−0.041, +0.016]
F 32 px + mask	A	+0.252 [+0.175, +0.343] $*$	+0.311 [+0.214, +0.422] $*$	+0.184 [+0.118, +0.281] $*$	+0.284 [+0.194, +0.382] $*$	+0.007 [−0.036, +0.046]
F 32 px + mask	D	+0.194 [+0.123, +0.282] $*$	+0.291 [+0.208, +0.384] $*$	+0.059 [0.000, +0.119]	+0.117 [+0.059, +0.182] $*$	−0.010 [−0.035, +0.011]

2.3 Detection by Defect Type (Final Configuration F)

Table 3: Detection by defect-type code for configuration F (102 defective images; recall as defined in Week 6, whose Table 11 gives the same table for the unmasked 64 px baseline).

Defect code	Images	Recall 1% FPR	Recall 5% FPR	Mean defect area (% of image)
019	38	0.184	0.579	0.003
030	13	0.462	0.615	0.015
002	9	0.333	0.889	0.100
010	9	1.000	1.000	2.300
022	9	0.333	0.667	0.029
006	8	1.000	1.000	0.191
025	5	0.200	1.000	0.002
023	5	0.800	1.000	0.087
016	3	1.000	1.000	0.078
027	1	1.000	1.000	0.027
029	1	0.000	0.000	0.029
036	1	1.000	1.000	22.268
All	102	0.451 (46/102)	0.745 (76/102)	–

2.4 Examples

The selection rules are those of Week 6 (fixed in advance, not by hand), now applied to configuration F.

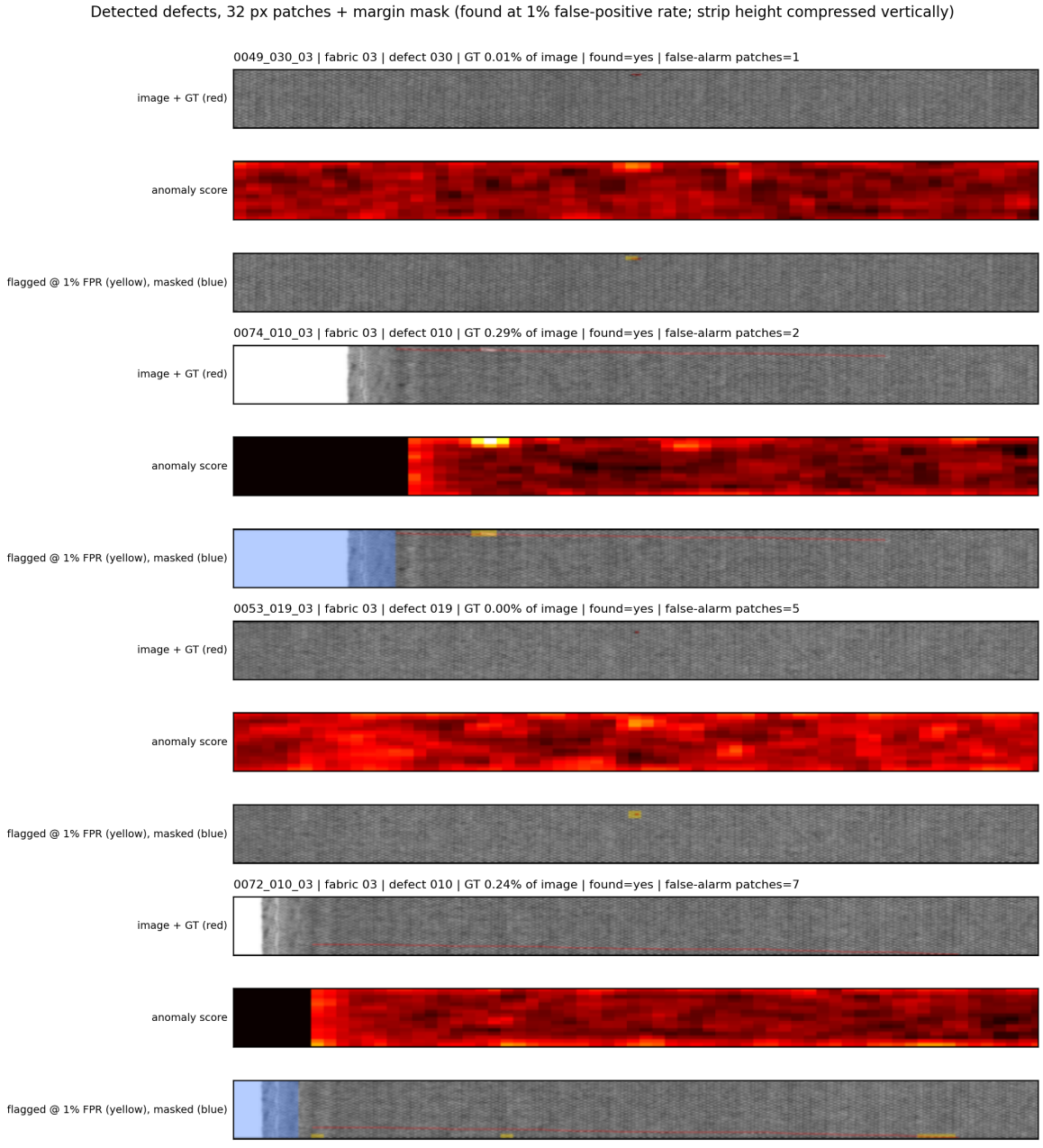


Figure 2: Detected defects with configuration F (found at 1% FPR). For each image: the image with the ground truth outlined in red, the anomaly score, and the flagged patches (yellow) with the masked margin zone (blue). The strip height is compressed vertically.

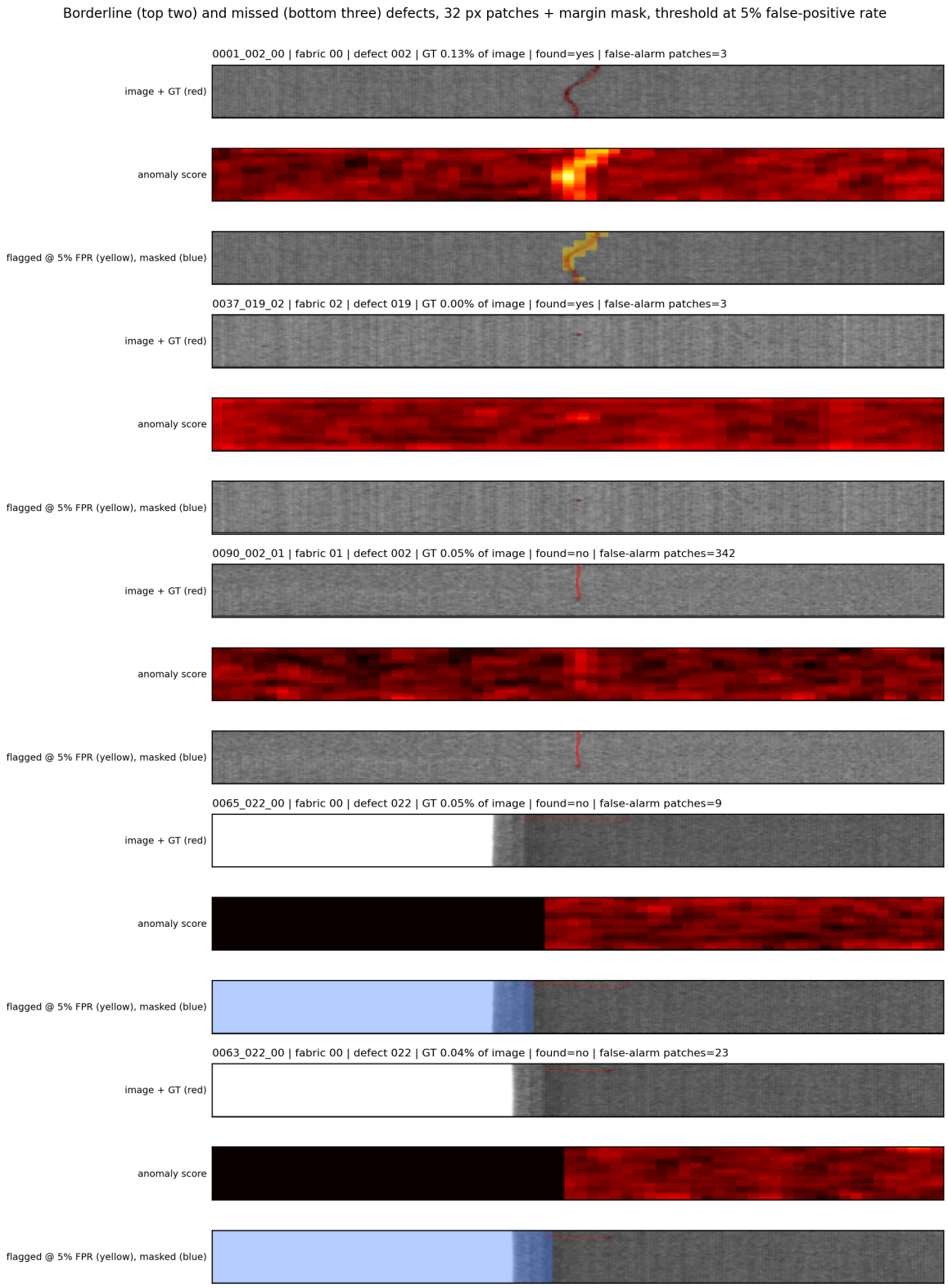


Figure 3: Borderline (found only at 5% FPR; top two) and missed (not found even at 5% FPR, largest defects; bottom three) defects with configuration F.

2.5 Interpretation

- **Final configuration: 32×32 patches with the corrected margin mask (F).** It finds the defect in 46 of 102 defective images at 1% FPR (0.451) and 76 of 102 at 5% FPR (0.745), against 20 and 44 for the baseline A. The top-1 patch overlaps the defect in 30.4% of images (A: 11.8%) and a top-40 patch in 51.0% (A: top-10, 22.5%). Against A the four defect-level and localization measures improve with intervals that exclude zero. Against the best 64 px configuration D, recall at 1% and 5% FPR and the top-10/40 hit rate improve reliably (top-1 borderline) and none gets reliably worse (Table 2), so under our decision rule F is adopted for localization.

- **Each change contributes something different.** Smaller patches mainly raise recall at 5% FPR (0.43 to 0.74 for E); the margin mask mainly helps localization (top-1 hit 0.167 to 0.304 from E to F, and higher point estimates for B and D than for A and C); the thin-defect features at 64 px do not change defect recall or localization (C against A) but raise the image-level AUC from 0.766 to 0.802 (+0.036 [+0.010, +0.064]).

- **Tiny defects are now found.** Defect type 019 (38 images, mean area 0.003% of the image, about 30 pixels) is found in 18.4% of images at 1% FPR and 57.9% at 5%, against 0% and 13% for the 64 px baseline in Week 6. In the detected examples, specks covering 0.00–0.01% of the image are flagged with one to five false-alarm patches, and thin lines are flagged along their length (Figure 2). Types 010, 006, 016, 027 and 036 are found in all images; type 029 (one image) is not found.

- **It is still not exact detection.** 56 of 102 defective images (55%) are missed at 1% FPR and 26 (25%) at 5%. The misses (Figure 3) are of two kinds: images whose whole score level is high (one missed image has 342 false-alarm patches at 5% FPR although its score peaks at the thin vertical defect), and thin lines at the boundary of the masked margin, which the mask partly removes.

- **Image-level detection stays modest and uses a different best configuration.** AUC is 0.772 for F, the same as for A, while the 64 px thin-defect features reach 0.802. Localization and whole-strip detection therefore do not share a best configuration.

- **Limitations.** The caveats of Week 6 still apply (lenient "found" criterion, thresholds estimated from defect-free patches defined with the ground-truth masks, operating points read on the same defect-free images). New this week: the 32 px configurations use one cross-validation seed, and their patch-level average precision varies too much between folds (0.253 ± 0.212 for F) to rank configurations, which is why we rank on the image-bootstrap measures; and the first margin mask removed white defects because the check that exposed it was not part of our protocol.

3 Work To Be Done

- Repeat the 32 px configurations with several cross-validation seeds and run the paired tests of B and D against A after the mask correction.

- Tighten the "found" criterion (for example require a flagged patch near the defect and limit the number of flagged patches per image) and re-evaluate the final configuration.

- Reduce the two remaining kinds of misses: images with a high background score level (for example a per-image score normalization judged by the localization measures rather than strict labels), and thin defects at the margin boundary (a less aggressive edge zone).

- Prepare the final report and the presentation around the six-configuration comparison and the example figures.
- Carried over from Week 6 and still open: re-run the post-processing and preprocessing experiments on the frozen sample; select image-level thresholds without ground-truth masks and post-processing on a validation split; follow up on illumination flattening; identify the 14 four-channel images; name the 12 defect types from the AFID paper; correct the earlier fabric-ID and defect-code statements in the final report.

4 Literature Review

No change since the Week 6 report; the reference list below is unchanged.

5 Challenges Faced

- Our first margin mask also removed bright (white) defects. It was found only through the example figures and an implausible masking statistic (236 of 244 images masked); a run-based margin definition fixed it. Checks of this kind are needed whenever a new filtering step is added.
- Comparing patch sizes is delicate: at the same false-positive rate there are about 4.3 times as many false-alarm patches at 32 px, so we compare false-alarm area, recall and localization rather than counts.
- Even with 32 px patches, defects of a few dozen pixels remain at the limit of what patch texture statistics can show: the final configuration still misses 55% of the defective images at 1% FPR and 25% at 5%.
- Per-fabric models and thresholds leave few defect-free images per fabric (20 or 21), so the threshold for each fabric is estimated from limited data.

References

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